

Sohyla Saeed Abdelrahim

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Summary

Computer Science student with a strong foundation in building robust, full-stack applications, now focused on applying those skills to design and deploy scalable data pipelines and machine learning models. Passionate about leveraging engineering principles to solve complex data problems and deliver data-driven products.

Education

Bachelor's degree in Computer Science – Faculty of Computers and Artificial Intelligence, Cairo University.
Cumulative GPA: 3.91/4.00 (2022 – 2026)

Graduation Project

Talk2System | *AI-Powered Requirements Engineering Platform* [GitHub](#) | 2025 – 2026

- Built a full-stack AI-driven platform that automates the software development lifecycle from raw stakeholder conversations to production-ready deliverables.
- Engineered an NLP preprocessing pipeline for transcript cleaning, semantic deduplication, and requirement extraction.
- Designed and trained machine learning classifiers for sentence-level multi-class classification of transcript segments into software engineering categories, achieving **96% accuracy**
- Implemented scalable LLM processing with adaptive chunking and fallback orchestration using **Ollama (Qwen 7B)** and **Google Gemini APIs** to support transcripts exceeding 80K words.
- Developed async background processing, role-aware dashboards, and requirement approval/versioning workflows.
- **Tech stack:** FastAPI, PostgreSQL, scikit-learn, spaCy, FastCoref, Ollama, Gemini API, React

Work Experience

Data Science Trainee — Digital Egyptian Pioneers Initiative (IBM Curriculum) July 2025 – Dec 2025

- Analyzed and preprocessed large real-world datasets using **Pandas** and **NumPy**, improving data quality for model training by handling missing values and outliers.
- Created visualizations and dashboards with **Matplotlib**, **Seaborn**, and **Streamlit** to communicate findings and model performance to a non-technical audience.
- Built, trained, and evaluated supervised and unsupervised **machine learning models** using **Scikit-learn** for classification and regression tasks.

Database Administration Intern – e& Egypt Aug. 2025 – Sept. 2025
Cairo, Egypt

- Gained hands-on knowledge in **Oracle Database Administration and Architecture**.
- Completed the **Oracle Database 19c Administration Workshop**, covering installation, configuration, and management.
- Performed practical tasks related to database creation, user management, backup, and recovery as part of the workshop.

Projects

Quiz Management System (Full stack) [GitHub](#)

- Developed a full-stack web application using **Django**, **HTML**, **CSS**, **JS**, and **DRF** enabling teachers to create timed quizzes and students to take them with auto-evaluation and result reporting.

Sales Forecasting and Optimization Project (Data Science) – Team Project [GitHub](#)

- Forecasted weekly sales for retail stores using time-series models **XGBoost**, **LightGBM**, **LSTM**, achieving accuracy **98%**.
- Performed **EDA**, **interactive Dashboards** and **Model deployment**

Material Stream Identification System (Machine Learning) [GitHub](#)

- Developed an Automated Material Stream Identification (MSI) System using fundamental Machine Learning (ML) techniques.
- Performed Data Augmentation, Feature Extraction, Model development and deployment achieving accuracy of **89%** using **SVM**.

Learning Management System (Backend) – Team Project [GitHub](#)

- Collaborated with a team of 5 to develop core backend services for an educational platform using **Java Spring Boot**. Implemented assignments submission and grading mechanism.

Netflix Show Clustering (Machine Learning)

[GitHub](#)

- Developed a Netflix show clustering and recommendation system using unsupervised machine learning.
- Designed and implemented data preprocessing, feature engineering, and **KMeans** clustering to group shows by similarity.
- Built a **Streamlit** web app and CLI tools for interactive recommendations, allowing users to find similar shows or predict clusters for new titles.

Smart Appointment Booking System (Backend)

[GitHub](#)

- Developed a **Spring Boot and SQL** backend for an intelligent scheduling system that optimizes appointment booking using real-time data.

Event Booking System (Full stack)

[GitHub](#)

- Built a full-stack web app for browsing, booking, and managing events with an integrated admin panel using **Django, HTML, CSS, and JS**.

Certifications & Courses

Oracle Database 19c Administration Workshop – **e& Egypt**

SQL Fundamentals Track – DataCamp

Associate Data Analyst in SQL – DataCamp

The Complete 2024 Web Development Bootcamp – Udemy

Fundamentals of Backend Engineering – Udemy

GitHub Foundations – DataCamp

Technical Skills

Languages: C++, Python (Pandas, NumPy, Scikit-learn, Matplotlib), Java, PHP, JavaScript, SQL

Frameworks & Libraries: Django, Django REST Framework (DRF), Spring Boot, Laravel

Databases & Tools: PostgreSQL, MySQL, Oracle 19c, Git, GitHub

Concepts: Machine Learning, Data Analysis, OOP, Data Structures, Algorithms, Database Administration, NLP, Data Warehousing.

Soft Skills

Collaboration & Teamwork: Proven ability to work effectively in team environments on multiple projects.

Adaptability: Quick learner with experience across multiple tech stacks (Java/Spring, Python/Django, PHP/Laravel) and domains (Web Dev, Data Science, DBA).

Time Management: Successfully balances a demanding academic workload (3.91 GPA) with project development and internship responsibilities.

Extracurricular Activities

Google Developer Student Club (GDSC) – UI/UX Committee Member

(Feb 2024 – July 2024)

Microsoft Student Partner Tech Club (MSP)